

Riyansh Singh Bhadouriya

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Professional Summary

AI-focused Computer Science student experienced in Deep Learning, Transformers, Retrieval-Augmented Generation (RAG), and Explainable AI. Skilled in designing scalable AI systems and backend ML pipelines using PyTorch, FastAPI, FAISS, and OpenAI APIs for semantic retrieval and cybersecurity applications.

Education

Indian Institute of Information Technology, Design and Manufacturing Kancheepuram

B.Tech in Computer Science Engineering (Artificial Intelligence)

Expected April 2027

CGPA: 7.54

Rankers International, Indore

Class XII (CBSE)

March 2022

Score: 89.4%

Skills

Languages: Python, C++, JavaScript, SQL, C,

AI & Machine Learning: Deep Learning, Transformers, Computer Vision, NLP, Retrieval-Augmented Generation (RAG), Explainable AI (XAI), Semantic Search, Attention Mechanisms, CNNs, RNNs, LSTMs

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, MONAI, OpenCV, LangChain, LangGraph, NumPy, Pandas, Matplotlib

LLM & Backend Systems: FastAPI, Flask, REST APIs, FAISS, Vector Embeddings, Prompt Engineering, AI Agents, Context-Aware Prompting, LLM APIs

Deployment & Tools: Docker, AWS, Railway, CI/CD, Git, GitHub, Linux, Jupyter Notebook, VS Code, MySQL, MongoDB, MLOps

Projects

RAGboYT – AI-Powered YouTube RAG Chatbot Extension

Engineered a production-grade **Retrieval-Augmented Generation (RAG)** Chrome extension enabling timestamp-grounded conversational querying over YouTube videos using semantic retrieval and LLM orchestration. Designed a modular backend pipeline using **FastAPI**, **FAISS**, and OpenAI APIs for transcript ingestion, semantic chunking, vector embeddings, prompt augmentation, and grounded response generation. Built scalable REST APIs supporting keyword re-ranking, vector similarity retrieval, and context-aware semantic search workflows. Developed the frontend using React, TypeScript, Tailwind CSS, and Plasmio with evidence tracing, timestamp navigation, and Railway cloud deployment.

Agentic AI-Powered Brain Tumor Segmentation and Clinical Analysis System

Developed a multimodal brain tumor segmentation pipeline using **Attention U-Net**, hybrid deep learning architectures, and CLIP-based semantic representation learning on the BraTS2020 and TextBraTS datasets. Processed FLAIR, T1, T1CE, and T2 MRI modalities using PyTorch, MONAI, and OpenCV with Dice Loss, IoU evaluation, Grad-CAM, attention heatmaps, and LLM-based clinical reporting workflows. Achieved a **96.59% Dice score**, **93.47% IoU**, validation loss of 0.1012, and zero failure cases below 0.90 Dice.

Explainable Transformer-Based IIoT Intrusion Detection using Temporal Attention Learning

Designed and executed a transformer-based IIoT intrusion detection framework using the ToN-IoT and CICIoT2023 datasets, improving multiclass cyberattack detection accuracy from **24% to 94.29%** through temporal sequence modeling and attention-based learning. Implemented SHAP-based explainability, attention heatmap visualization, and comparative evaluation against Random Forest classifiers for feature-level attack attribution, temporal behavior analysis, and contextual sequence reasoning.